

(1)

ML, NLP, DL Bootcamp

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Quartiles

Variable:-

variable is a property

{ Age = 25, Gender = male

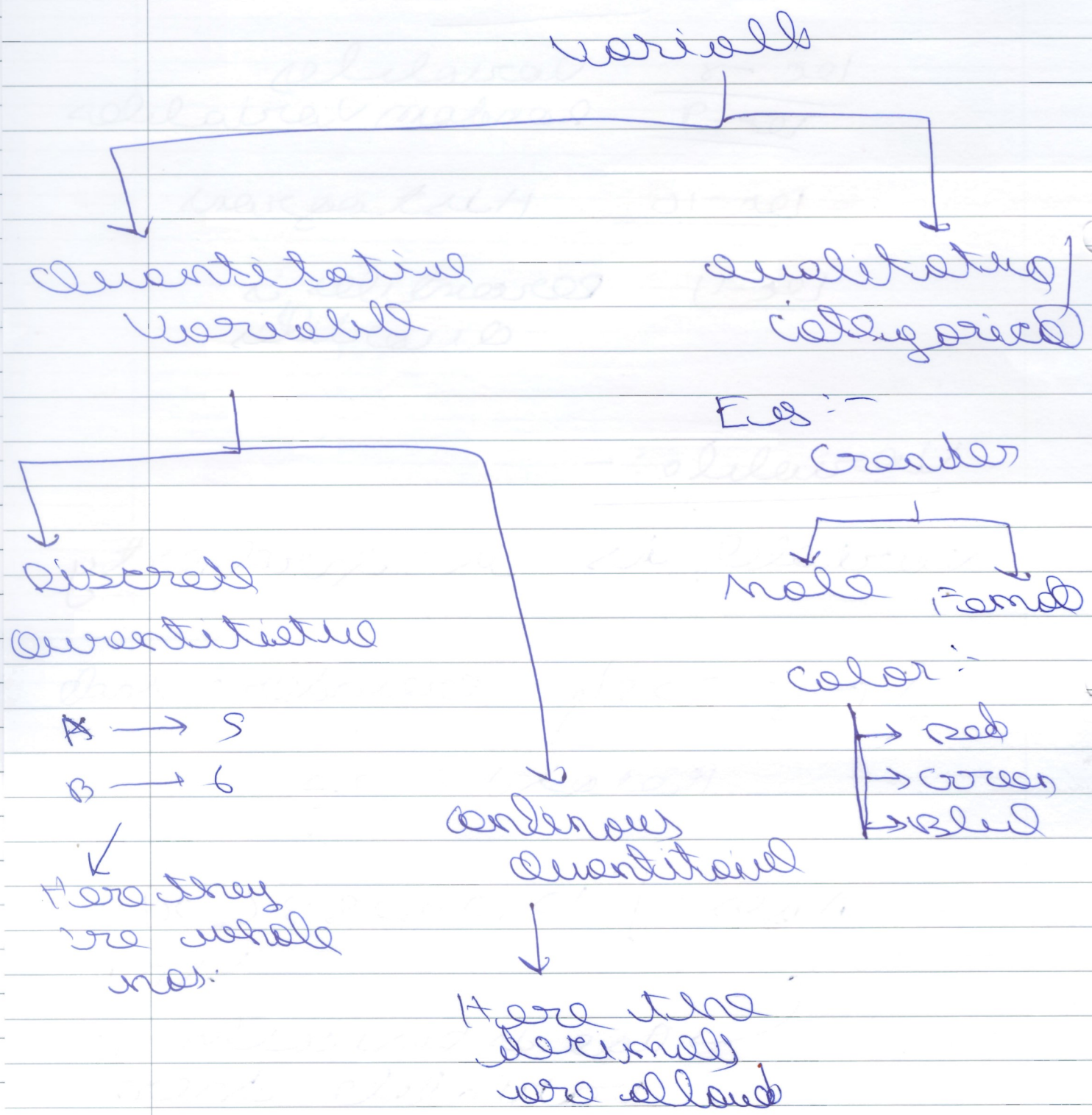
Height = 1.2

Ages = [12, 20, 25, 30, 70]

x-Variable

→ Age is basically a
variable here

=> Significant types of variables



⇒ Random Variables

Random variables are denoted by X .

(continuous)

Suppose an equation is given as

$$y = 5x + 2 \quad \text{let } x = 1$$

$$y = 5 + 2 \Rightarrow x = 2$$

$$x = 3$$

→ Here we assign different x values to y values.

Random variable is a function where values are derived from different functions or processes.

$X = \left\{ \begin{array}{l} \text{lossing a coin} \\ \text{possibilities are H \& T} \end{array} \right.$
 ↙
 (Heads)
 (Tails)

$X = \left\{ \begin{array}{l} 0 \quad H \\ 1 \quad T \end{array} \right.$

Suppose the random variables are three

$X = \left\{ \begin{array}{l} 1 \\ 2 \\ 3 \\ 4 \\ 5 \\ 6 \end{array} \right. \rightarrow \text{rolling of die}$

Types of Random

Discrete
random

Ex: Tossing a
coin

Rolling a
die

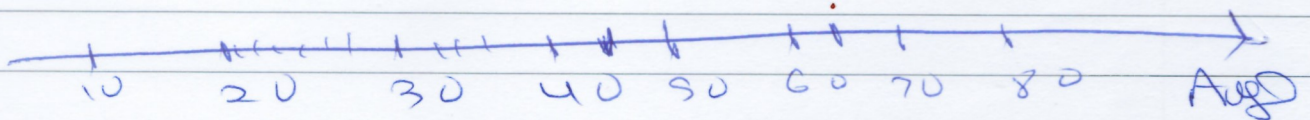
Continuous
Random
Variable

Ex: Tomorrow
how many
inches is
it going
to rain

=> Histograms:-
Ages

$$X = \{ 23, 24, 25, 30, 34, 36, 40, 50, \\ 60, 75, 80 \}$$

Suppose, we take all
the readings on x axis



(6)

Suppose, the ranges are as follows

$$20-30 = 4$$

$$30-40 = 3$$

$$40-50 = 2$$

$$50-60 = 1$$

$$60-70 = 0$$

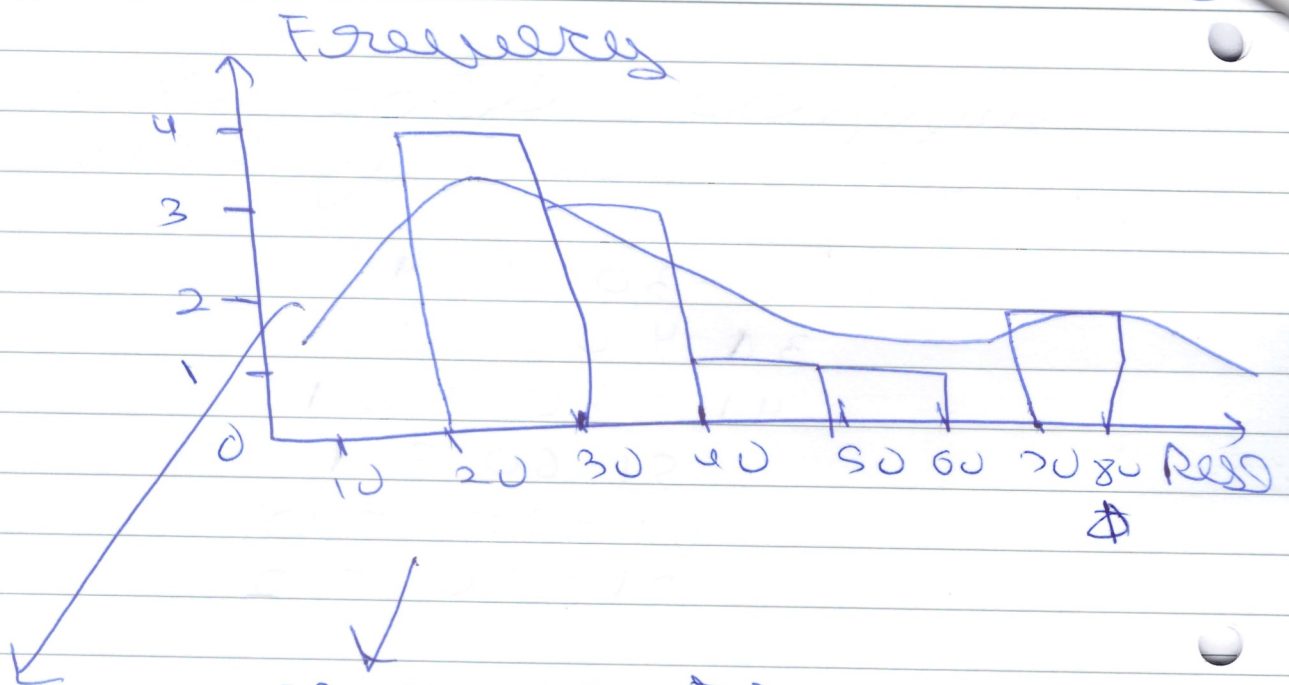
$$70-80 = 2$$

Histogram is basically counting of frequency based on bins.

Here the bin size = 10

(0-10), (10-20), (20-30), (30-40)
these are basically the bins of size 10.

Here the y axis will now become the frequency.



Q How to smooth the Histogram?

→ For this we use kernel Density Estimation (KDE)

Percentiles & Quartiles

$$\text{percentiles} = \{1, 2, 3, 4, 5, 6\}$$

$$\text{No. of odd numbers} = 3$$

$$\text{Percentage of odd numbers in group} = \frac{3}{6}$$

$$\frac{3}{6} \times 100 = 50\%$$

Percentile :-

A percentile is a value below which a certain percentage of observations lie

$$= \{2, 2, 3, 4, 5, 5, 6, 7, 8, 8, 8, 9, 9, 10\}$$

$$\text{Percentile of value below } x = 9 = \frac{\text{No. of values below } x}{n} \times 100$$

$$\text{No. of values below } 9 = 11$$

$$\frac{11}{14} \times 100 = 78.57\%$$

Hence the percentile
ranking = 78.571 of
values

⇒ Suppose now we
need the 25th percentile
from the data

then the formula

$$\text{val} = \frac{\text{percentile}}{100} \times (n+1)$$

$$n = 14$$

$$= \frac{25}{100} \times (14+1)$$

$$= \frac{1}{4} \times 15$$

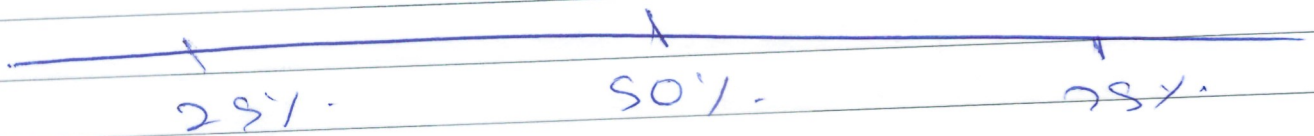
$$= 3.75 \text{ A.}$$

$$\approx 3.5$$

2

Now we round off
to around 3.5

=> Quartiles



quartile basically
means a quarter

25% = 1st quartile

50% = 2nd quartile

75% = 3rd quartile